

universität freiburg



CS
DEPARTMENT OF
COMPUTER SCIENCE

Test and Reliability

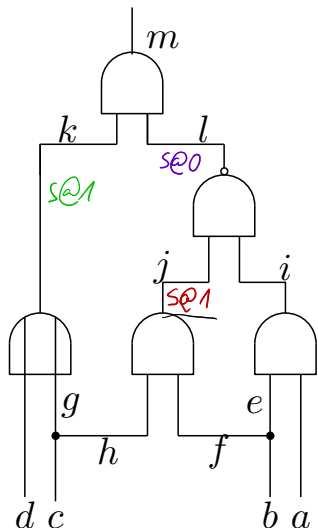
Exercise Sheet 2

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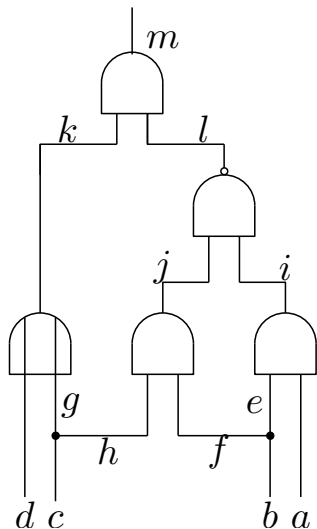


Exercise 1a) Fault Parallel Simulation



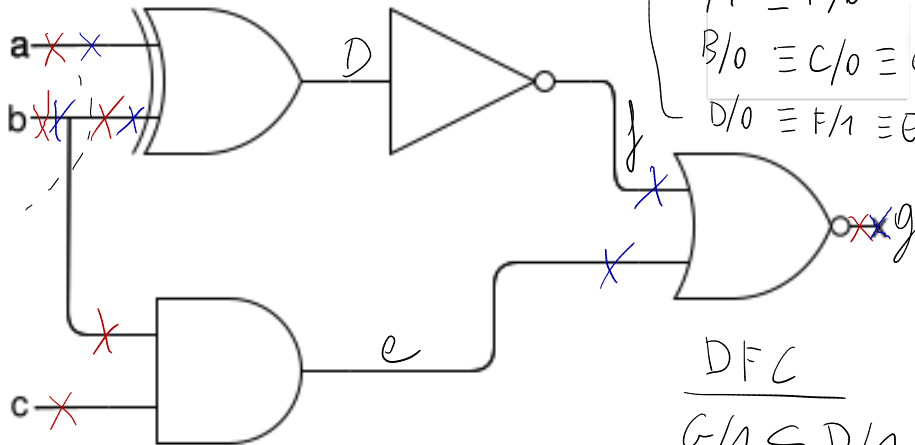
$a = 0$ $h = 0$
 $b = 1$ $i = 0$
 $c = 0$ $j = 0$ ✓
 $d = 1$ $k = 1$
 $e = 1$ $l = 1$ 0 ✓
 $f = 1$ $m = 1$ 0 ✓
 $g = 0$

Exercise 1a) *Fault* Parallel Simulation



	fault-free	j s.a. 1	l s.a. 0	k s.a. 1
a	0	0	0	0
b	1	1	1	1
c	0	0	0	0
d	1	1	1	1
e	1	1	1	1
f	1	1	1	1
g	0	0	0	0
h	0	0	0	0
i	0	0	0	0
j	0	1	0	0
k	1	1	1	0
l	1	1	0	1
m	1	1	0	1

Exercise 1b) EFC



EFC (intuition)

$$D/1 \equiv F/0$$

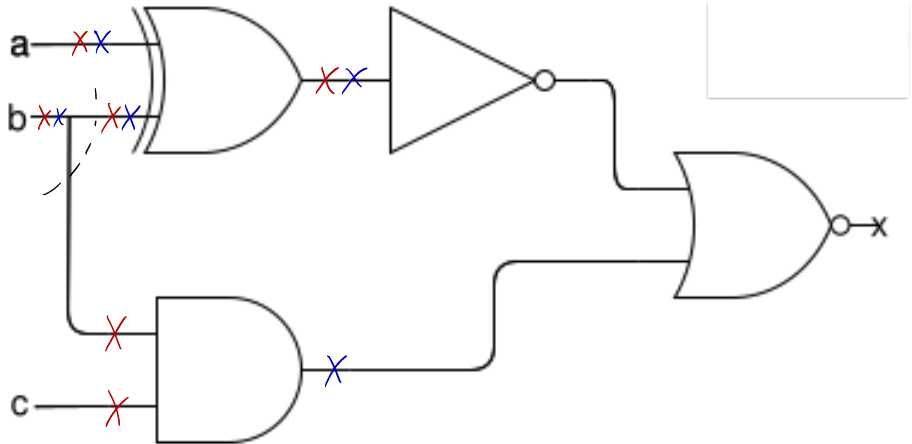
$$B/0 \equiv C/0 \equiv E/0$$

$$D/0 \equiv F/1 \equiv E/1 \equiv G/0$$

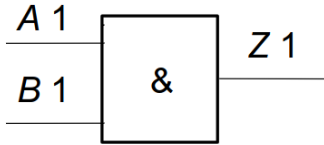
DFC

$$G/1 \subseteq D/1$$

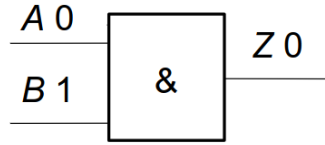
Exercise 1b) DFC



Exercise 2, Deductive Fault Simulation

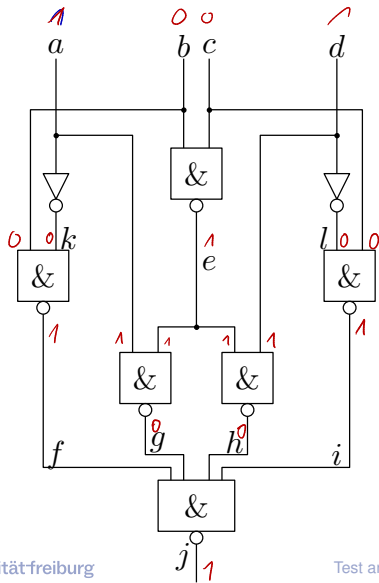


$$L_Z = L_A \cup L_B \cup \{Z/0\}$$

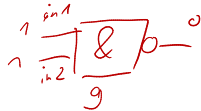


$$L_Z = (L_A - L_B) \cup \{Z/1\}$$

Exercise 2, pattern 1001



$$L_g = \{out/0\}$$



$$L_g = L_{in1} \cup L_{in2} \cup \{out/1\}$$

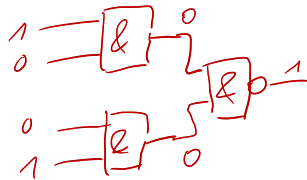
$$L_a = \{a/0\}, L_b = \{b/1\}, L_c = \{c/1\}, L_d = \{d/0\}$$

$$L_e = \{e/0\}, L_f = \{f/0\}, L_i = \{i/0\}$$

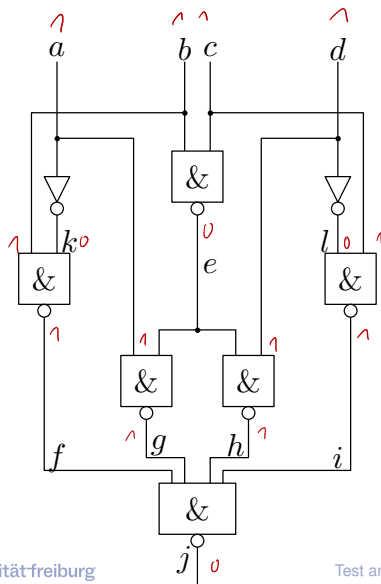
$$L_g = L_a \cup L_e \cup \{g/1\} = \{a/0, e/0, g/1\}$$

$$L_h = \{d/0, e/0, h/1\}$$

$$L_j = \{j/0\}$$



Exercise 2, pattern 1111



i either a, b, c, d

$$L_i = \{i/0\}$$

$$L_k = \{k/1\}$$

$$L_f = (L_k \setminus L_b) \cup \{f/0\} = \{k/1, f/0, a/0\}$$

$$L_i = \{i/1, i/0, d/0\}$$

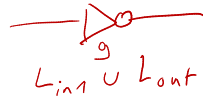
$$L_{ge} = \{b/0, c/0, e/1\}$$

$$L_g = \{b/0, c/0, e/1, g/0\}$$

$$L_h = \{b/0, c/0, e/1, h/0\}$$

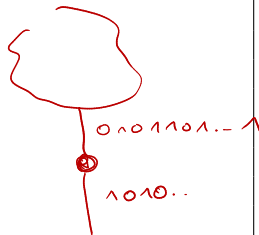
$$L_j = L_f \cup L_g \cup L_h \cup L_i \cup \{j/1\}$$

$$= \{k/1, f/0, b/0, c/0, e/1, g/0, h/0, i/1, i/0, j/1\}, \{a/0, d/0\}$$



Exercise 3

```
unsigned char deref(unsigned literal)
{
    unsigned int variable_of_literal = literal / 2;
    unsigned int result = current_valuation[variable_of_literal]
    unsigned int negated_literal = literal & 1;
    if (negated_literal)
    {
        // something is missing here, Hint: UINT_MAX = 11...111
        result = result ^ UINT_MAX;
    }
    return result;
}
```



Exercise 3

```
...
unsigned int l, r, tmp, j;
for (j = 0; j < model->num_and; j++) // iterate over all gates
{
    aiger_and *and = model->ands + j;
    l = deref (and->rhs0); // and->rhs0 is the first input literal of AND
    r = deref (and->rhs1); // and->rhs1 is the second input literal of AND
    tmp = l & r;
    current_valuation[and->lhs / 2] = tmp; // store output value (and->lhs)
}
```

